

## No. of Functional Features Included in the Solution

|                      |              |
|----------------------|--------------|
| <b>Date</b>          | 3 July 2026  |
| <b>Team ID</b>       | XXXXXX       |
| <b>Project Name</b>  | Smart Lender |
| <b>Maximum Marks</b> | 5 Marks      |

### Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

| S.No | Feature Name                 | Feature Description                                                                 | Module / Component  | Status | Marks Contribution |
|------|------------------------------|-------------------------------------------------------------------------------------|---------------------|--------|--------------------|
| 1    | User Registration & Login    | Allows authorized users to access the application securely.                         | Authentication      | Done   | High               |
| 2    | Applicant Data Entry         | Collects applicant details such as income, education, employment, loan amount, etc. | Frontend (Flask UI) | Done   | High               |
| 3    | Data Preprocessing           | Cleans and transforms user input before prediction.                                 | ML Pipeline         | Done   | High               |
| 4    | Loan Eligibility Prediction  | Predicts loan approval using the trained XGBoost model.                             | Prediction Engine   | Done   | High               |
| 5    | Real-Time Prediction         | Displays instant loan approval/rejection results.                                   | Flask Backend       | Done   | High               |
| 6    | Input Validation             | Validates user inputs and prevents invalid data submission.                         | Validation Module   | Done   | Medium             |
| 7    | Model Performance Evaluation | Compares Decision Tree, Random Forest, KNN, and XGBoost models.                     | Model Evaluation    | Done   | Medium             |
| 8    | IBM Cloud Deployment Ready   | Application prepared for deployment on IBM Cloud.                                   | Deployment          | Done   | Medium             |

Feature Summary:

| Metric                             | Count / Value                                                                   |
|------------------------------------|---------------------------------------------------------------------------------|
| Total Features Planned             | 8                                                                               |
| Total Features Implemented         | 8                                                                               |
| Core / Must-Have Features          | Loan Prediction, Applicant Data Entry, Data Preprocessing, XGBoost Prediction   |
| Additional / Nice-to-Have Features | Input Validation, Model Comparison, IBM Cloud Deployment Support, Responsive UI |
| Features Tested & Verified         | All 8 Features                                                                  |

Feature Category Breakdown:

| S.No | Category                  | Features in Category | Example Features                                   |
|------|---------------------------|----------------------|----------------------------------------------------|
| 1    | User Interface (UI)       | 2                    | Applicant Form, Prediction Result Page             |
| 2    | Backend / Logic           | 3                    | Flask Routing, Data Preprocessing, Loan Prediction |
| 3    | Database / Storage        | 1                    | Trained Model (.pkl) Storage                       |
| 4    | API / Integration         | 1                    | Flask Prediction API                               |
| 5    | Security / Authentication | 1                    | Input Validation and Secure Form Handling          |